

from (i) & (ii)

$$2EA = \frac{\sigma A}{\epsilon_0}$$

$$\# \boxed{E = \frac{\sigma}{2\epsilon_0}}$$

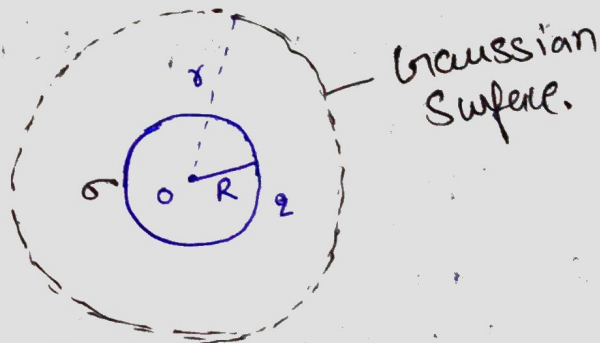
* E is independent of x

(3) field due to a uniformly charged thin spherical shell.

Let σ be the uniform surface charge density of a thin spherical shell of radius R .

(i) field outside the shell.

Consider a point P outside the shell with radius vector r . To calculate E at P we take the Gaussian surface.



E and Δs at every point are parallel &

$$\phi = E \Delta s$$

$$\phi = E 4\pi r^2 \quad \text{--- (i)}$$

$$\phi = \frac{1}{\epsilon_0} q_{en}$$

$$\sigma = \frac{q}{4\pi R^2}$$

$$\phi = \frac{\sigma 4\pi R^2}{\epsilon_0} \quad \text{--- (ii)}$$